

Series GEFH1/C



SET ~ 3

रोल नं.

Roll No.



प्रश्न-पत्र कोड
Q.P. Code

55/C/3

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book. *

भौतिक विज्ञान (सैद्धान्तिक)

PHYSICS (Theory)

निर्धारित समय : 3 घण्टे

अधिकतम अंक : 70

Time allowed : 3 hours

Maximum Marks : 70

नोट / NOTE :

- (i) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 27 हैं।
Please check that this question paper contains 27 printed pages.
- (ii) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- (iii) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 35 प्रश्न हैं।
Please check that this question paper contains 35 questions.
- (iv) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
Please write down the serial number of the question in the answer-book before attempting it.
- (v) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.



General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) This question paper contains **35** questions. **All** questions are **compulsory**.
- (ii) This question paper is divided into **five** Sections – **A, B, C, D** and **E**.
- (iii) In **Section A** – Questions no. **1** to **18** are Multiple Choice (MCQ) type questions, carrying **1** mark each.
- (iv) In **Section B** – Questions no. **19** to **25** are Very Short Answer (VSA) type questions, carrying **2** marks each.
- (v) In **Section C** – Questions no. **26** to **30** are Short Answer (SA) type questions, carrying **3** marks each.
- (vi) In **Section D** – Questions no. **31** to **33** are Long Answer (LA) type questions carrying **5** marks each.
- (vii) In **Section E** – Questions no. **34** and **35** are case-based questions carrying **4** marks each.
- (viii) There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 3 questions in Section D and 2 questions in Section E.
- (ix) Use of calculators is **not** allowed.

Use the following values of physical constants, if required :

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\text{Mass of electron (m}_e\text{)} = 9.1 \times 10^{-31} \text{ kg}$$

$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

$$\text{Boltzmann constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}$$

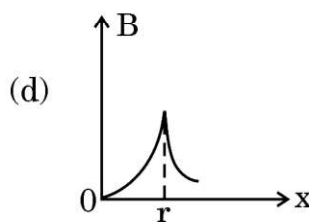
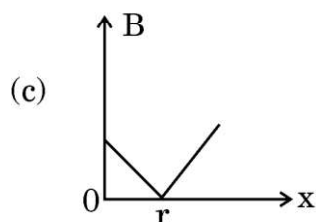
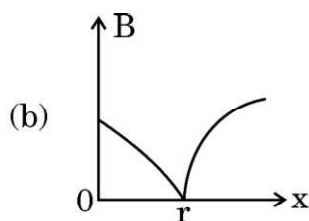
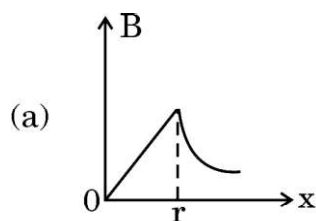


SECTION A

1. An electric dipole with dipole moment $\vec{P} = P_0 \hat{i} - P_0 \hat{j}$ is placed in an electric field $\vec{E} = E_1 \hat{i} + E_2 \hat{j}$, where P_0 , E_1 and E_2 are constants. The torque $\vec{\tau}$ acting on the dipole is :

- (a) $P_0 (E_2 - E_1) \hat{k}$ (b) $P_0 (E_2 + E_1) \hat{k}$
 (c) $-P_0 (E_2 + E_1) \hat{k}$ (d) $P_0 (E_1 - E_2) \hat{k}$

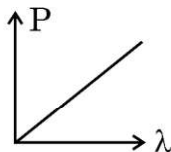
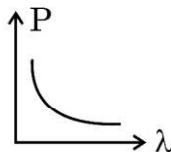
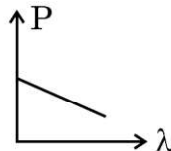
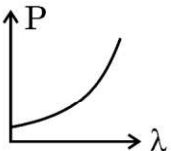
2. The current I flowing through a conductor of radius r is uniformly distributed across its cross-section. Which of the following graphs represents the variation of magnitude of magnetic field $\vec{B}$ with distance x from the axis of the conductor ?

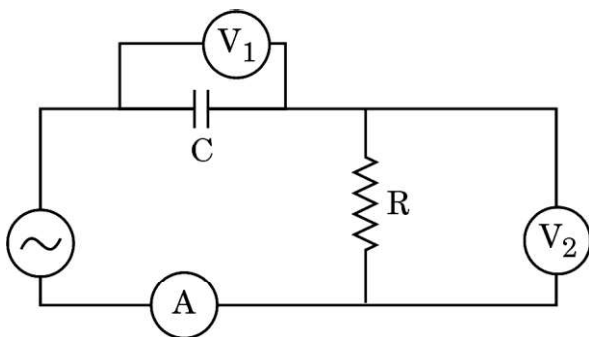
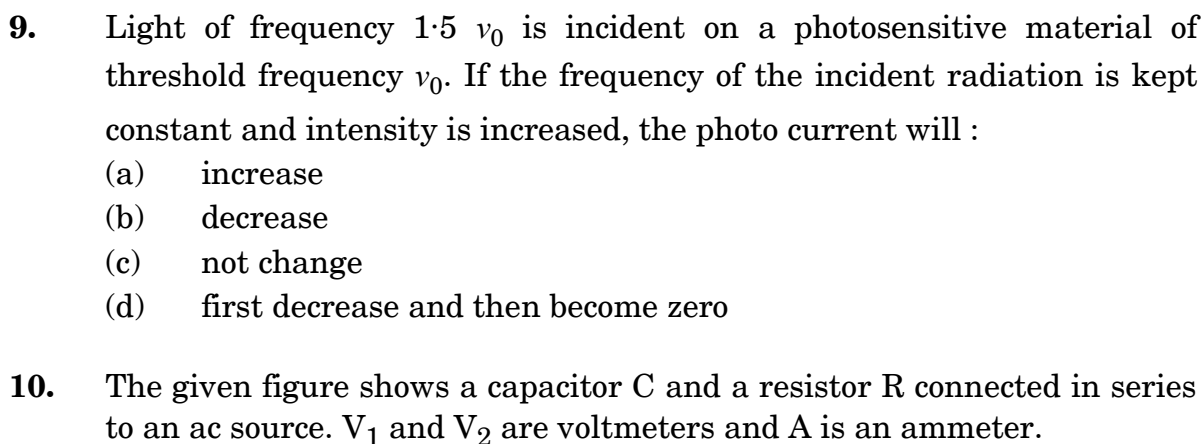


3. A bar magnet of pole strength (m) and magnetic moment (M) is cut perpendicular to its axis in two equal halves. The new pole strength (m') and magnetic moment (M') of each part are respectively :

- (a) m and M (b) m and $\frac{M}{2}$
 (c) $\frac{m}{2}$ and $2M$ (d) $2m$ and $\frac{M}{2}$



4. In a series LC circuit connected to an ac source, with the increase in the frequency of the source, the net reactance :
- (a) increases linearly
 - (b) decreases linearly
 - (c) first increases to become maximum and then decreases to zero
 - (d) first decreases to become zero and then increases
5. Which of the following radiations has the highest frequency ?
- (a) Visible light
 - (b) Infrared rays
 - (c) Microwaves
 - (d) X-rays
6. In Young's double-slit experiment, the intensity on the screen is I_0 at a point where path difference is λ . The intensity at the point where path difference is $\frac{\lambda}{4}$ is :
- (a) $\frac{I_0}{4}$
 - (b) $\frac{I_0}{2}$
 - (c) I_0
 - (d) zero
7. Which of the following figures represents the variation of a particle's momentum with the de Broglie wavelength associated with it ?
- (a) 
- (b) 
- (c) 
- (d) 
8. The binding energy per nucleon is lower :
- (a) for light nuclei only
 - (b) for heavy nuclei only
 - (c) for nuclei of middle mass numbers only
 - (d) both for the light nuclei and the heavy nuclei



Which of the following statements is correct ?

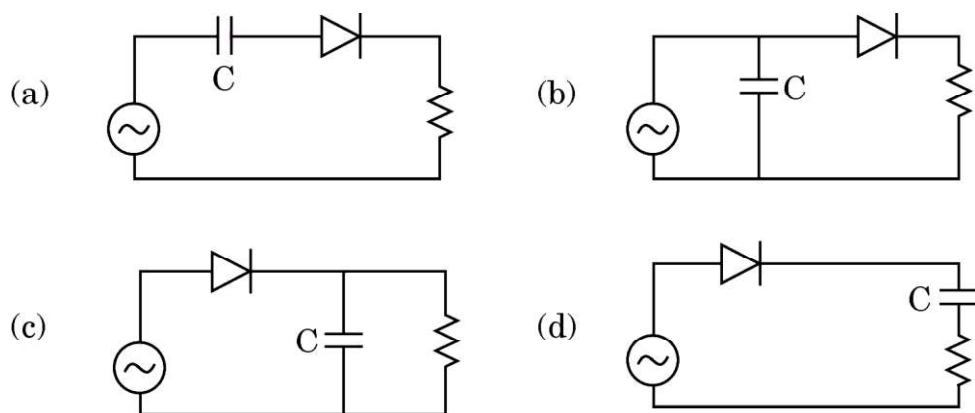
- (a) Current in the circuit lags in phase with voltage shown in V_2 .
- (b) The voltage shown in V_1 is ahead in phase with voltage shown in V_2 .
- (c) The current in the circuit and the voltage shown in V_1 are always in phase.
- (d) The voltage shown in V_1 lags behind in phase with the voltage shown in V_2 .
- 11.** Two nuclei have their mass numbers in the ratio of 1 : 27. What is the ratio of their nuclear densities ?
- (a) 1 : 27 (b) 1 : 1
- (c) 1 : 9 (d) 1 : 3
- 12.** The impact parameter for an alpha particle approaching a target nucleus is maximum when the scattering angle (θ) is :
- (a) 0° (b) 90°
- (c) 180° (d) 45°



13. A planar loop is rotated in a magnetic field about an axis perpendicular to the field. The polarity of induced emf changes once in each :

- (a) 1 revolution (b) $\left(\frac{1}{2}\right)$ revolution
(c) $\left(\frac{1}{4}\right)$ revolution (d) $\left(\frac{3}{4}\right)$ revolution

14. In which of the following diagrams is the capacitor 'C' connected correctly to provide smooth output of a half-wave rectifier ?



15. A plane wave is incident on a concave mirror of radius of curvature R. The reflected wave is a spherical wave of radius :

- (a) $\frac{R}{4}$ (b) $\frac{R}{2}$
(c) R (d) 2R

Questions number 16 to 18 are Assertion (A) and Reason (R) type questions. Two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false and Reason (R) is also false.



16. *Assertion (A)* : The temperature coefficient of resistance is positive for metals and negative for semi-conductors.
Reason (R) : The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.
17. *Assertion (A)* : A proton and an alpha particle having same kinetic energy are moving in circular paths in a uniform magnetic field. The radii of their circular paths will be equal.
Reason (R) : The centripetal force required to move a charged particle in a circle does not depend on the magnitude of the magnetic field.
18. *Assertion (A)* : Silicon is preferred over germanium for making semiconductor devices.
Reason (R) : The energy gap for germanium is more than the energy gap for silicon.

SECTION B

19. (a) The electric field of an electromagnetic wave passing through vacuum is represented as $E_x = E_0 \sin (kz - \omega t)$. Identify the parameter which is related to the (i) wavelength, and (ii) the frequency of the wave in the above equation.
(b) Write two properties of a medium that determine the velocity of light in that medium. 2
20. Two identical bars, one of a paramagnetic material and another of a diamagnetic material are kept in a uniform magnetic field. Show diagrammatically the modifications in the pattern of magnetic field in each case. How are the two materials affected by increase in temperature ? 2
21. Light of wavelength $3500 \text{ \AA}$ is incident on two metals A and B. Which of them will yield photoelectrons, if their work functions are 4.2 eV and 1.9 eV respectively ? Make the necessary calculations to justify your answer. 2



22. A uniform electric field is represented as $\vec{E} = (3 \times 10^3 \frac{N}{C}) \hat{i}$. Find the electric flux of this field through a square of side 10 cm when the :
- (a) plane of the square is parallel to y-z plane, and
 - (b) the normal to plane of the square makes an angle of 60° with the x-axis.

2

23. A beam of high energy photons, each of energy of E is incident on a metal surface and electrons are ejected from the surface. Assuming that the work function of the metal, ϕ is negligible, find the relation between the de Broglie wavelength λ associated with photoelectrons and the energy of photons E. Show in a figure, the variation of λ as a function of E.

2

24. (a) A uniform electric field E of 500 N/C is directed along +x axis. O, B and A are three points in the field having x and y coordinates (in cm) (0, 0), (4, 0) and (0, 3) respectively. Calculate the potential difference between the points (i) O and A, and (ii) O and B.

2

OR

- (b) Three point charges $1 \mu C$, $-1 \mu C$ and $2 \mu C$ are kept at the vertices A, B and C respectively of an equilateral triangle of side 1 m. A_1 , B_1 and C_1 are the midpoints of the sides AB, BC and CA respectively. Calculate the net amount of work done in displacing the charge from A to A_1 , from B to B_1 and from C to C_1 .

2

25. (a) When is more power delivered to a light bulb — just after it is turned on and the glow of the filament is increasing or after the glow becomes steady ? Why ?

2

OR

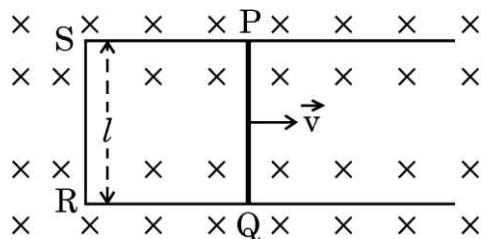
- (b) A battery is connected first across the series combination and then across the parallel combination, of three resistances R, 2R and 3R. In which of the three resistances will power dissipated be maximum in the two cases ? Justify your answer.

2



SECTION C

26. The figure shows a rectangular conductor PQRS with a movable arm PQ, kept in a uniform magnetic field $\vec{B}$ pointing into the page.



- (i) PQ is moved towards the right with a velocity $\vec{v}$. Obtain the expression for the emf developed across PQ.
 - (ii) If r is the resistance of PQRS, find the force required to move PQ with constant velocity $\vec{v}$.
- 3
27. (a) (i) An electron in a hydrogen atom jumps from second excited state to the first excited state. Name the spectral series in the spectrum of hydrogen atom to which the emitted radiation belongs.
- (ii) Find the ratio of the wavelengths of the “most energetic spectral” lines in the Balmer series to that in Paschen series of Hydrogen spectrum.
- 3

OR

- (b) (i) An α -particle having kinetic energy K approaches a nucleus of atomic number Z . It gets close to the nucleus and then approaches a distance (d) and reverses its direction. Obtain an expression for the distance of closest approach (d) in terms of kinetic energy of the α -particle.
- (ii) A proton and an alpha particle approach a target nucleus in head-on position, with equal velocities. Find the ratio of their distances of closest approach to the target nucleus.
- 3



- 28.** (a) (i) In which case is diffraction effect more dominant — slit formed by 2 blades or slit formed by two fingers ?
- (ii) Yellow light ($\lambda = 6000 \text{ \AA}$) illuminates a single slit of width $1 \times 10^{-4} \text{ m}$. Calculate (i) the distance between two dark lines on either side of central maximum, in the diffraction pattern observed on a screen kept 1.5 m away from the slit, and (ii) the angular spread of the first minimum. 3

OR

- (b) (i) What will be the colour of the central bright fringe in Young's double slit experiment if the monochromatic source is replaced by a source of white light ? Give reason for your answer.
- (ii) In Young's double slit experiment, the slits are separated by 0.3 mm and the screen is placed 1.5 m away from the slits. The distance between the central bright fringe and the sixth bright fringe is found to be 1.8 cm. Find the wavelength of light used in the experiment. 3

- 29.** (a) It is not advisable to use a galvanometer as such to measure current directly. Why ?
- (b) Why should the value of resistance connected in parallel to a galvanometer be low ?
- (c) Is the reading shown by an ammeter in a circuit less than or more than the actual value of current flowing in the circuit ? Why ? 3

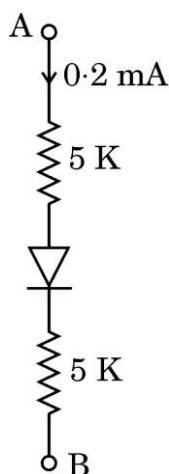
- 30.** Answer the following, giving reasons : 3
- (a) Induced emf does not always produce induced current.
- (b) The motion of a copper plate is damped when it is allowed to oscillate between pole pieces of a strong magnet.
- (c) No power is consumed in an ac circuit containing a pure inductor.



SECTION D

31. (a) (i) Draw the circuit diagram used to study I – V characteristics of a p-n junction diode in conducting mode. Mark on the graph the threshold voltage of the diode. Explain the significance of this voltage.
- (ii) In the circuit shown in the figure, the forward voltage drop across the diode is 0.3 V. Find the voltage difference between A and B.

5



OR

- (b) (i) Briefly describe the classification of solids into metals, insulators and semi-conductors on the basis of energy level diagrams.
- (ii) In a silicon diode, the current increases from 10 mA to 20 mA when the voltage changes from 0.6 V to 0.7 V. Calculate the dynamic resistance of the diode.

5

32. (a) (i) Derive the relation between the current and the drift velocity of free electrons in a conductor. Briefly explain the variation of resistance of a conductor with rise in temperature.
- (ii) An ammeter, together with an unknown resistance in series is connected across two identical batteries, each of emf 1.5 V, connected (i) in series, and (ii) in parallel. If the current recorded in the two cases be $\left(\frac{1}{2}\right)$ A and $\left(\frac{1}{3}\right)$ A respectively, calculate the internal resistance of each battery.

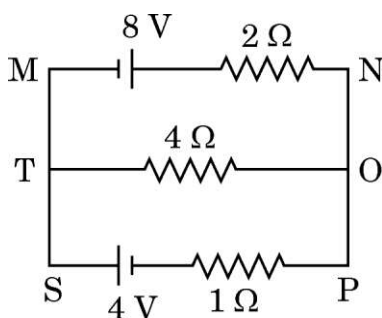
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OR



- (b) (i) State Kirchhoff's rules. Use them to obtain the condition of balance for a Wheatstone Bridge.
- (ii) Use Kirchhoff's rule to determine the currents flowing through the branches MN, TO and SP in the circuit shown in the figure.

5



33. (a) (i) Draw a ray diagram to show the formation of image by a concave mirror when the object is placed between f and $2f$. Using this diagram, derive the mirror equation.
- (ii) An object is kept 30 cm in front of a convex mirror of focal length 12 cm. Find the nature and position of the image formed.

5

OR

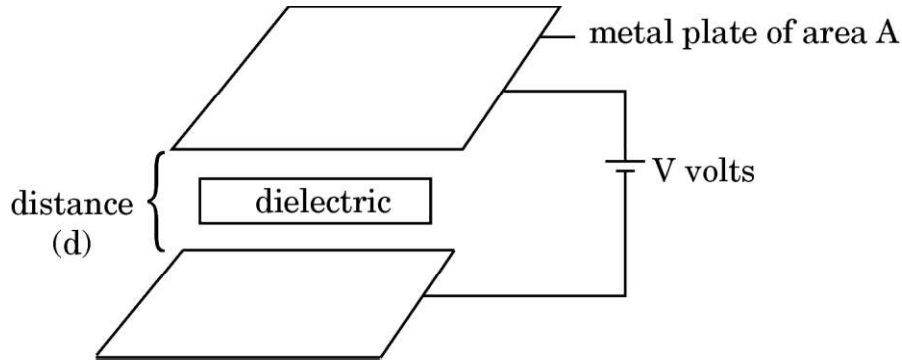
- (b) (i) Draw a ray diagram showing the image formation by a compound microscope. Hence obtain the expression for total magnification when the image is formed at least distance of distinct vision.
- (ii) A compound microscope consists of an objective lens of focal length 2.0 cm and an eyepiece of focal length 6.0 cm. If they are separated by a distance of 24 cm, find the total magnification when the image is formed at infinity.

5



SECTION E

34.



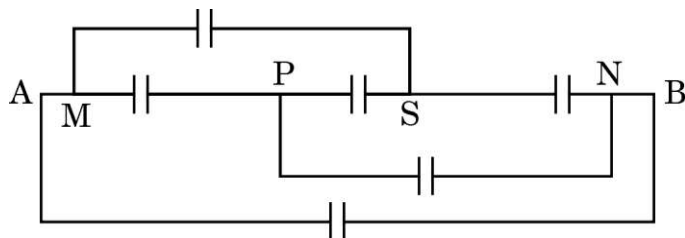
A parallel plate capacitor is an arrangement of two identical metal plates kept parallel, a small distance apart. The capacitance of a capacitor depends on the size and separation of the two plates and also on the dielectric constant of the medium between the plates. Like resistors, capacitors can also be arranged in series or parallel or a combination of both. By virtue of electric field between the plates, charged capacitors store energy.

- (a) The capacitance of a parallel plate capacitor increases from $10\ \mu\text{F}$ to $80\ \mu\text{F}$ on introducing a dielectric medium between the plates. Find the dielectric constant of the medium. 1
- (b) n capacitors, each of capacitance C , are connected in series. Find the equivalent capacitance of the combination. 1
- (c) A capacitor is charged to a potential (V) by connecting it to a battery. After some time, the battery is disconnected and a dielectric is introduced between the plates. How will the potential difference between the plates, and the energy stored in it be affected ? Justify your answer. 2

OR



- (c) Find the equivalent capacitance between points A and B, if capacitance of each capacitor is C. 2



35. A prism is a solid transparent medium bounded by three rectangular faces with triangular base and top. A ray of light incident at angle i on one face of a prism suffers two refractions on passing through a prism. Hence it deviates through a certain angle δ from its original path. The angle of deviation becomes minimum ($\delta = \delta_m$) for a certain value of angle i . In such a condition, the refracted ray inside the prism becomes parallel to its base. An expression for refractive index μ of the material of the prism can be obtained in terms of angle A and angle δ_m .

- (a) Show in a figure the variation of angle δ with angle of incidence i . 1
- (b) Show that for a prism of small angle A, the refractive index μ of its material can be written as $\mu = 1 + \frac{\delta_m}{A}$. 1
- (c) A ray of light passes through an equilateral prism such that both the angle of incidence and the angle of emergence are equal to the angle of prism A. Find the refractive index of the material of the prism, in terms of A. 2

OR

- (c) A ray of light passes through a prism of angle 75° , as shown in the figure. The refractive index of the material of the prism, with respect to its surrounding is $\sqrt{2}$. Find the angle of incidence i . 2

